



2019

ممثل

قسم هندسة الطاقة الكهربائية

مصطفى القرشي

اسئلة سنوات الكترونيات ١



طلاب البور والاتصالات



spark team



spark-fet.com



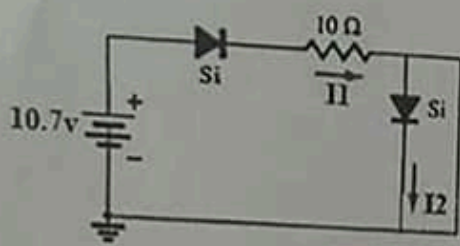


Fig.1

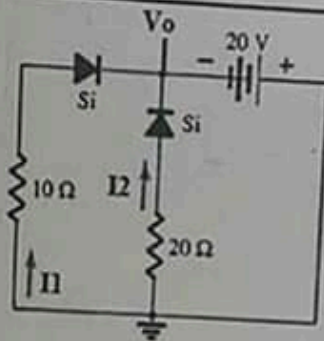


Fig.2

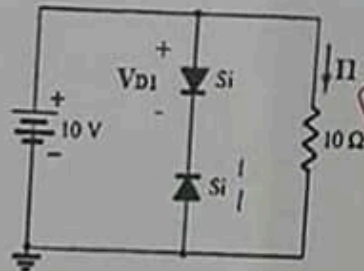


Fig.3

Q.1 With reference to figure 1, answer the next questions:

A- The value of the current I_1 is: 1 A

B- The value of the current I_2 is: 0 A "zero"

Q.2 With reference to figure 2, answer the next questions:

A- The value of the current I_1 is: 1.93 A

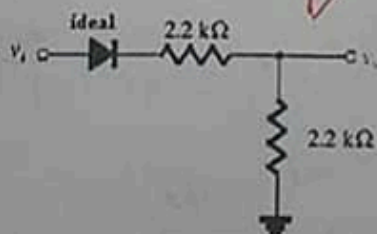
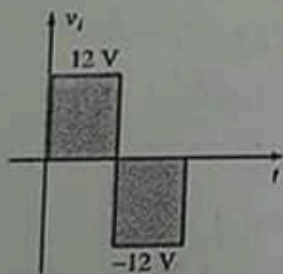
B- The value of the current I_2 is: 0.965 A

C- The value of the voltage V_o is: -20 V

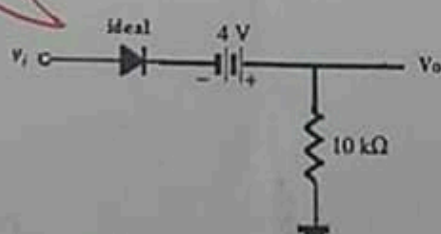
Q.3 With reference to figure 3, answer the next questions:

A- The value of the current I_1 is: 1 A

B- The value of the voltage V_{DI} is: 10 V → zero



(a)



(b)

Q.4 With reference to the above figure (a) and (b) sketch the output for each circuit, and determine the peak value on the sketch.

v_1

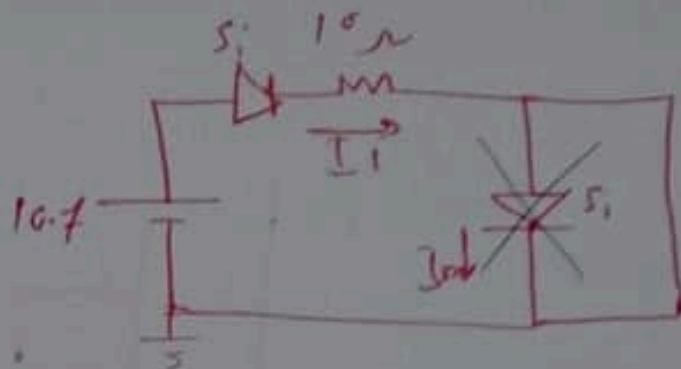
$V_{o1} = 6.999 \approx 7V$
 $V_{o2} = 0V$

$V_{o1} = 8V$

$V_{o2} = 0$

v_1

Q1



(A)

$$-10 + 0 + 10I_1 = 0$$

$$I_1 = 1A$$

$$B = I_2 = 2V = 0$$

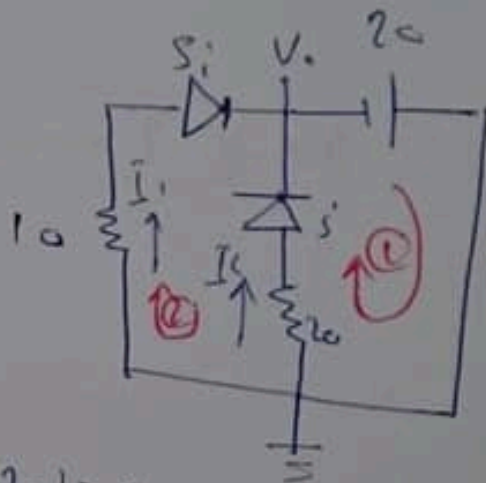
Short Circuit

Q2

(C)

$$20I_2 + 0.7 - 20 = 0$$

$$I_2 = 0.965A$$



(2)

$$I_1(10) + 0.7 - 0.7 - I_2(20) = 0$$

$$I_1 = 1.93A$$

$$V_0 = -20V$$

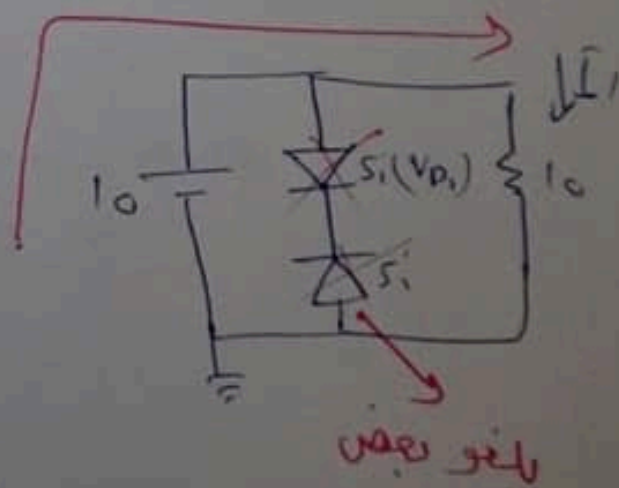
پول پتانسیل

Q3

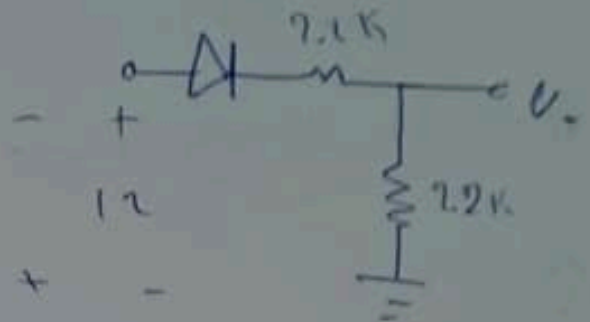
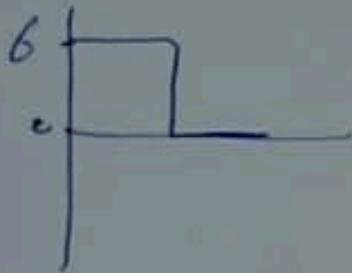
$$-10 + 10I_1 = 0$$

$$I_1 = 1A$$

$$V_{D1} = 2V$$



Q4

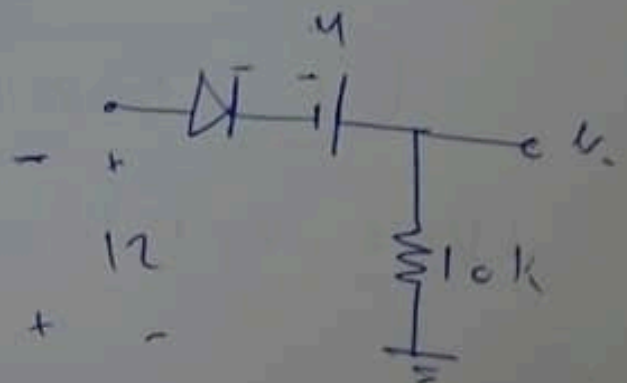
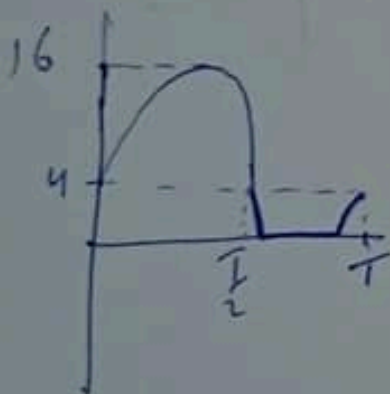


$V_o = 0 \text{ V}$

~~12V~~

$$V_o = \frac{12 \times 2.2k}{2.2k + 2.2k}$$

$V_o = 6 \text{ volt}$

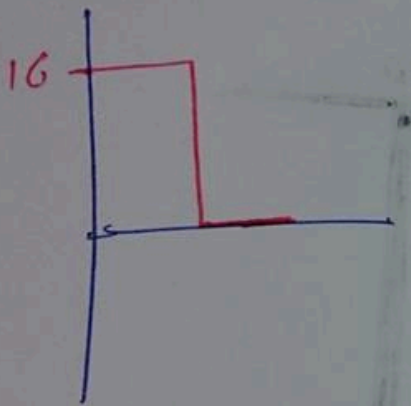


~~12V~~

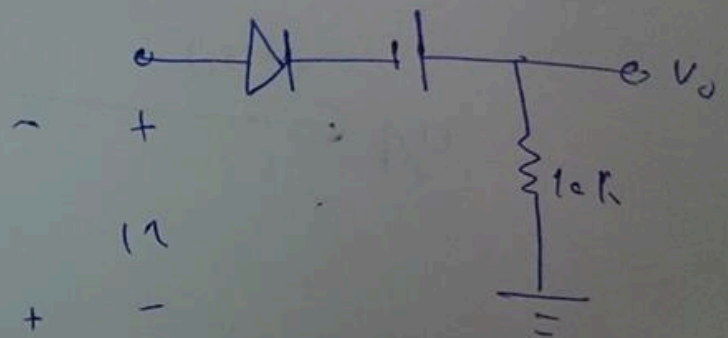
$V_o = 0 \text{ V}$

$$-12 - 4 + V_o = 0$$

$V_o = 16 \text{ volt}$



$$V_o = 20V_o$$



$$-16 - 4 + V_o = 0$$

$$V_o = 20V_o$$



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First Exam / Electronics I

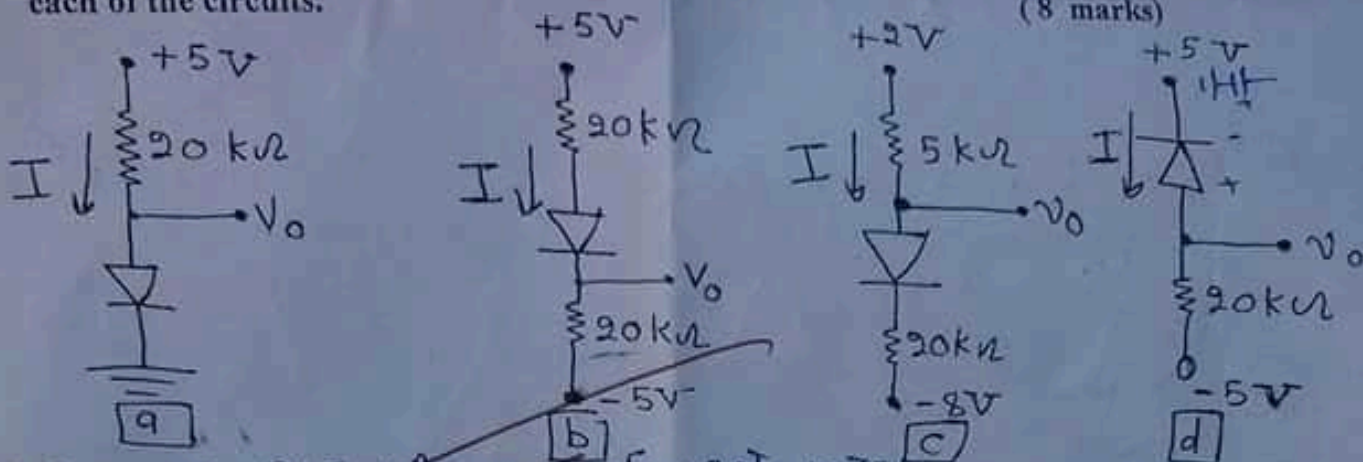


Instructors: D. Amjad Al-hindy, E. Khawla Omar

Name:

Date: Thursday 5/11/2015

Q1) The diode voltage is $V_D = 0.7\text{ V}$ in the four following circuits. Find I & V_o in each of the circuits. (8 marks)



a) $I = \dots 0.215\text{ mA}$ $-5 + 20I + V_o = 0$

$V_o = \dots 0.7\text{ V}$ $-5 + 20I + V_o = 0$

b) $I = \dots 0.2325\text{ mA}$ $-5 + 20I + 0.7 + 20I - 5 = 0$

$V_o = \dots -14.965\text{ V}$ $-V_o + 0.7 + 20I - 8 = 0$

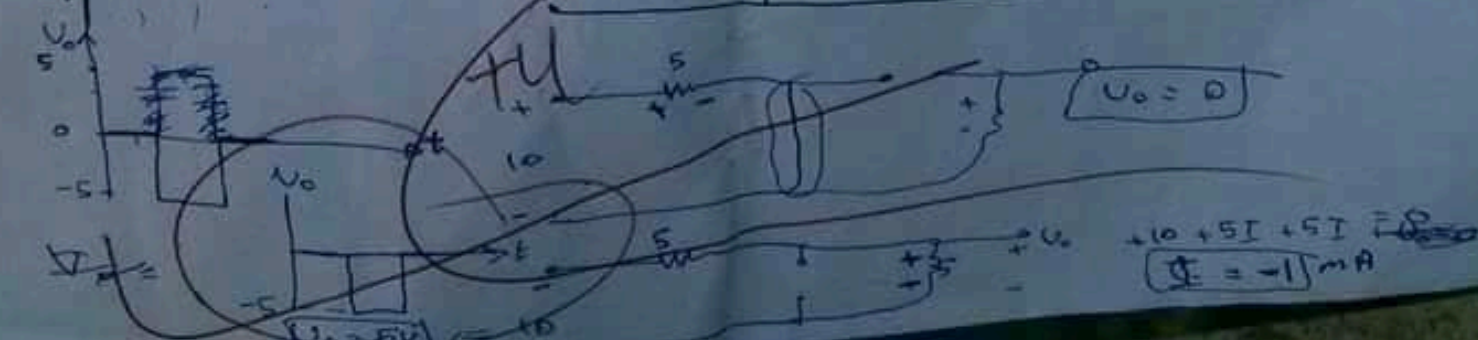
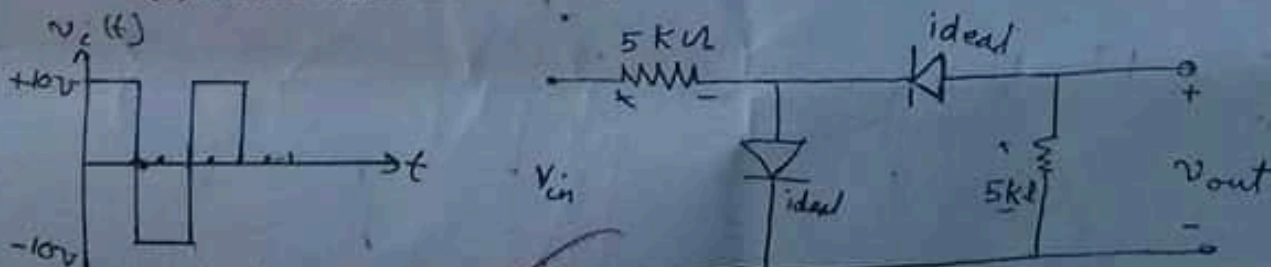
c) $I = \dots 0.372\text{ mA}$ $-2 + 5I + 0.7 + 20I - 8 = 0$

$V_o = \dots 0.14\text{ V}$ $-V_o + 0.7 + 20I - 8 = 0$

d) $I = \dots 0$ $I = 0$

$V_o = \dots -5$ $-V_o + 20 \times 0 - 5 = 0 \Rightarrow V_o = -5\text{ V}$

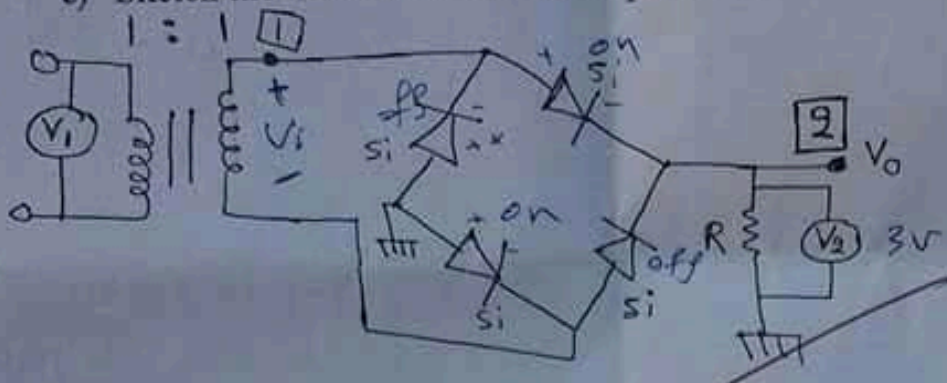
Q2) Sketch V_{out} for the following circuit. (4 marks)



Q3) If you know that the voltmeter (V_2) read 3V, $f_i = 50\text{Hz}$. Find:

(8 marks)

- The value that (V_1) will read.
- The output frequency, f_o .
- Sketch the waves with its value at points 1,2.



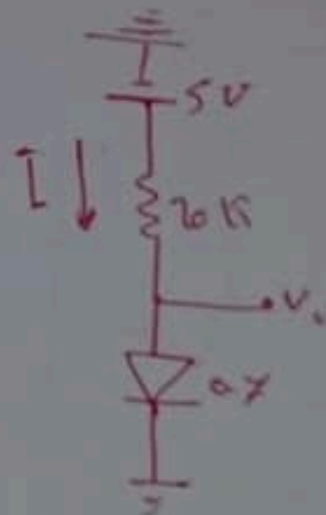
Q1 a

$$-5 + (20K)\bar{I} + 0.7 = 0$$

$$\bar{I} = 0.215 \text{ mA}$$

$$-5 + (20K)\bar{I} + V_o = 0$$

$$V_o = 0.7 \text{ Volt}$$



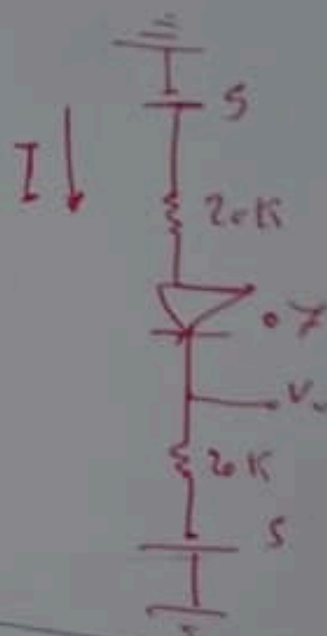
b

$$-5 + (40K)\bar{I} + 0.7 - 5 = 0$$

$$\bar{I} = 0.2325 \text{ mA}$$

$$5 - \bar{I}(20K) + V_o = 0$$

$$V_o = -0.35 \text{ Volt}$$



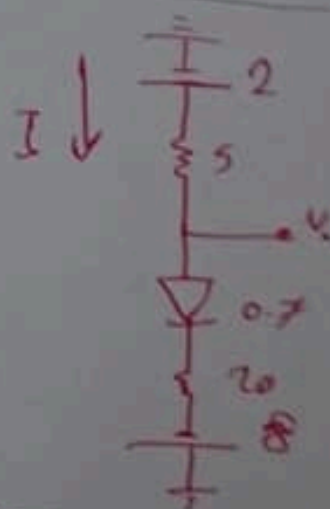
c

$$-2 + (25K)\bar{I} + 0.7 - 8 = 0$$

$$\bar{I} = 0.372 \text{ mA}$$

$$-2 + (5K)\bar{I} + V_o = 0$$

$$V_o = 0.14 \text{ Volt}$$



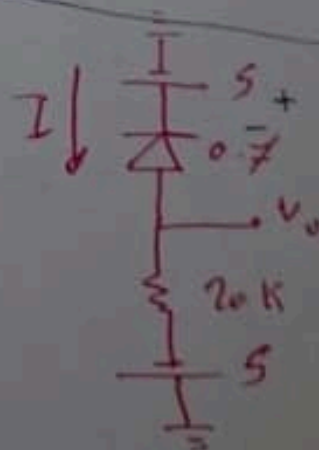
d

$$-5 + 2 + (20K)\bar{I} - 5 = 0$$

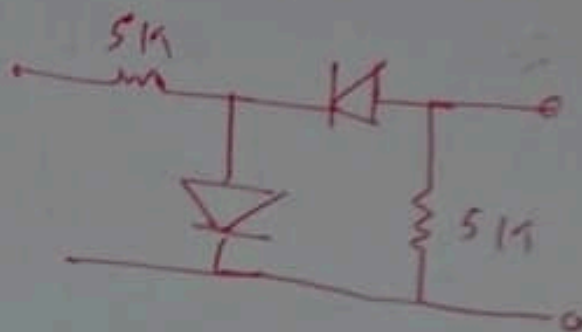
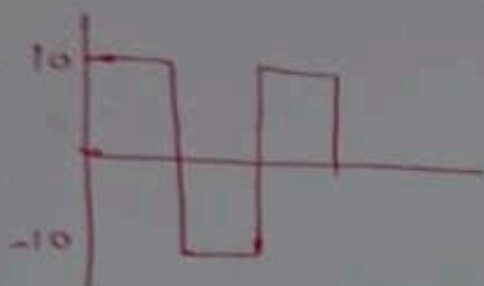
$\bar{I} =$

$$\bar{I} = 2 \text{ zero}$$

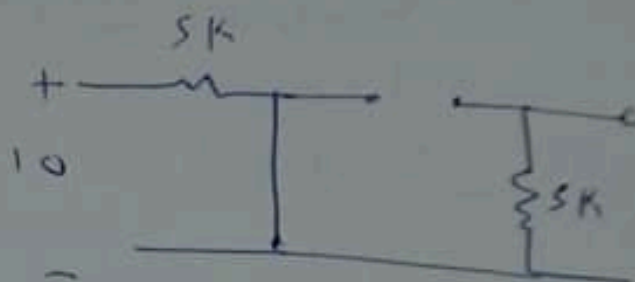
$$V_o = -5 \text{ Volt}$$



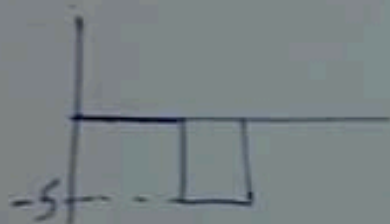
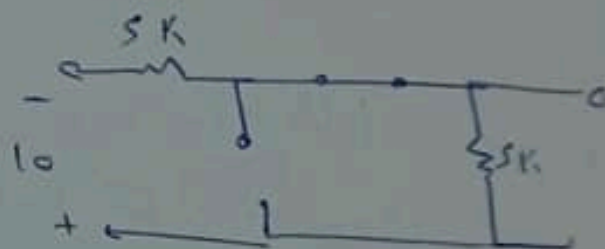
Q2



$$V_o = 2 \text{ V}$$



$$V_o = \frac{-10 \times 5k}{10k} = -5 \text{ V}$$



Q3

$$V_{DC} = 3 \text{ V} \rightarrow N_{DC} = \frac{2}{\pi} V_o$$

$$V_o = \frac{3\pi}{2} = 4.712 \text{ V}$$

(a)

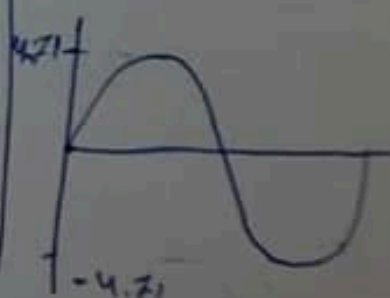
$$V_o = V_m - 2V_T \Rightarrow V_m = V_o + 2V_T$$

$$V_m = 6.112 \text{ V}$$

$$V_i = \frac{V_m}{R} = 4.32 \text{ V}$$

$$(b) f_{out} = 2 f_{in} = 2(50) = 100 \text{ Hz}$$

(c)



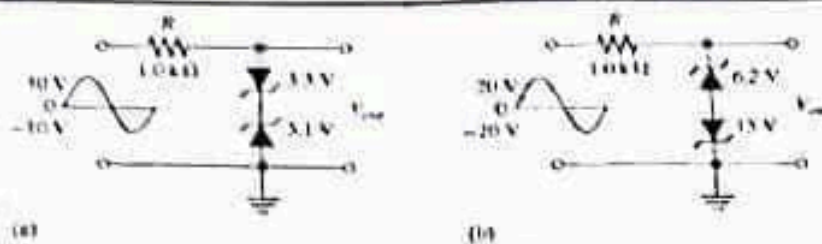


Fig.1

Q.1 A- Referring to figure 1-(a), answer the next questions (assuming that the zener diode in forward acts as a silicon diode):

1. The value of the output voltage at positive half cycle is: _____
2. The maximum current at negative half cycle is: _____

B- Referring to figure 1-(b), answer the next questions (assuming that the zener diode in forward acts as a silicon diode):

1. The value of the output voltage at negative half cycle is: _____
2. The maximum current at positive half cycle is: _____

Q.2 Sketch the output waveform and values, for each case in figures below (assuming that all diodes are ideal):

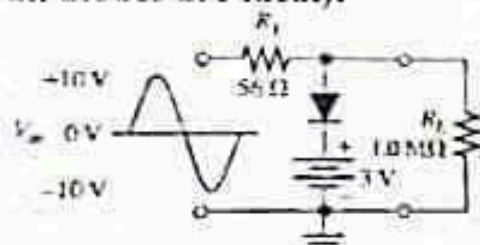


Fig.2

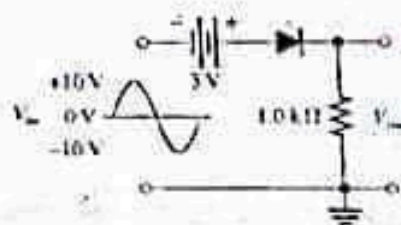


Fig.3

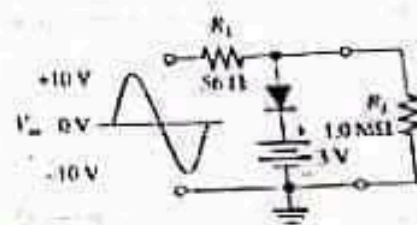


Fig.4

Fig.2

Fig.3

Fig.4

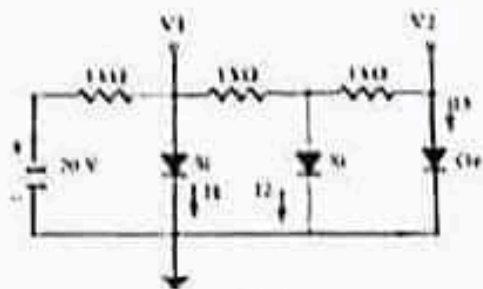


Fig.5

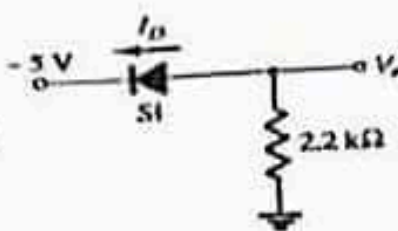


Fig.6

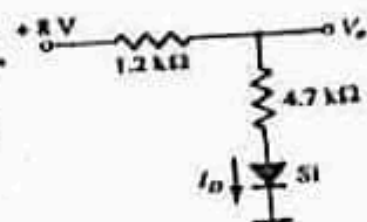


Fig.7

Q.5 With reference to figure 5, answer the next questions:

A- The value of the current I_1 is: _____

B- The value of the current I_2 is: _____

C- The value of the current I_3 is: _____

D- The value of the voltage V_1 is: _____

Q.6 With reference to figure 6, answer the next questions:

A- The value of the current I_1 is: _____

B- The value of the output voltage is: _____

Q.7 With reference to figure 7, answer the next questions:

A- The value of the current I_1 is: _____

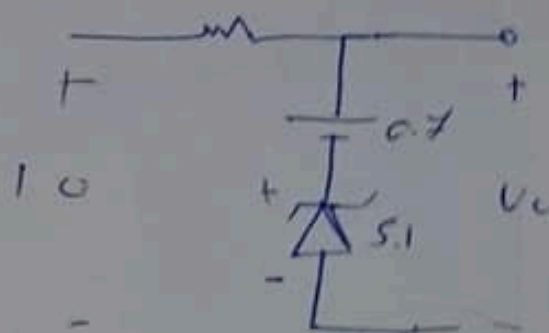
B- The value of the output voltage is: _____

Q.8 Design a voltage regulator circuit, that gives us an output voltage of 10v, choose any suitable values and draw the circuit below with all values on it.

Q1 (A)

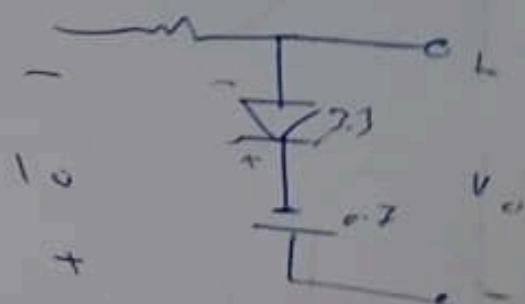
$$\textcircled{1} \quad -5.1 - 0.7 + V_o = 0$$

$$V_o = 5.8 \text{ volt}$$



$$\textcircled{2} \quad +0.7 + 3.3 + V_o = 0$$

$$V_o = -4 \text{ volt}$$



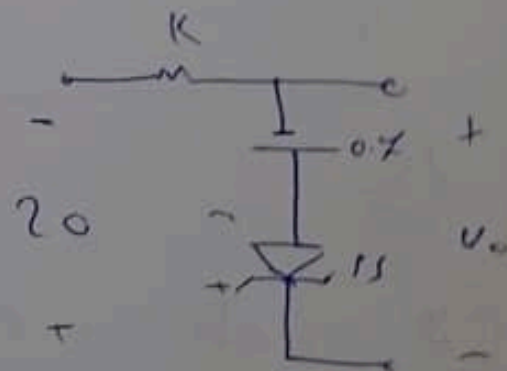
التيار في الدارة هو

$$I_{\text{mer}} = \frac{V_o}{1K} = -4 \text{ mA}$$

(B) (1)

$$15 + 0.7 + V_o = 0$$

$$V_o = -15.7 \text{ volt}$$

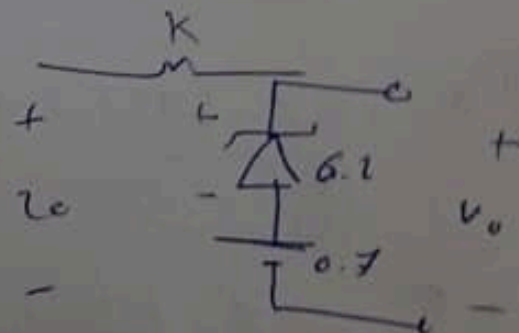


(2)

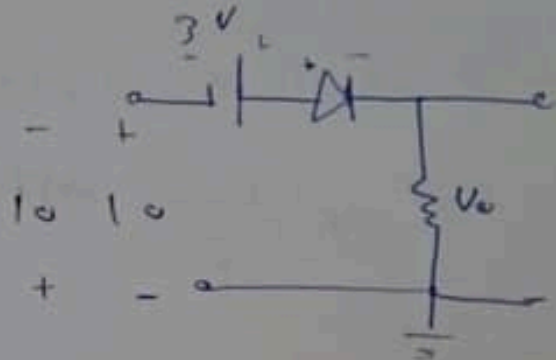
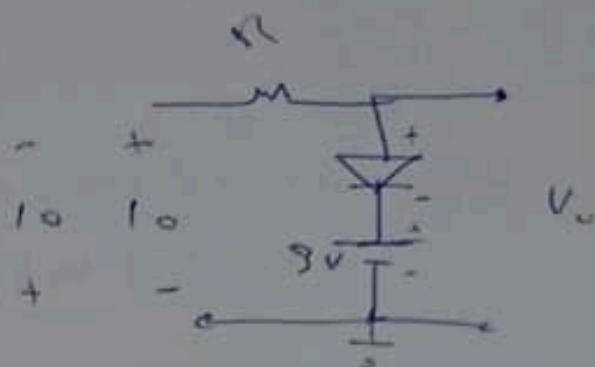
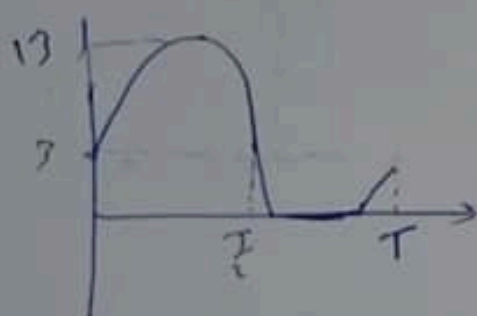
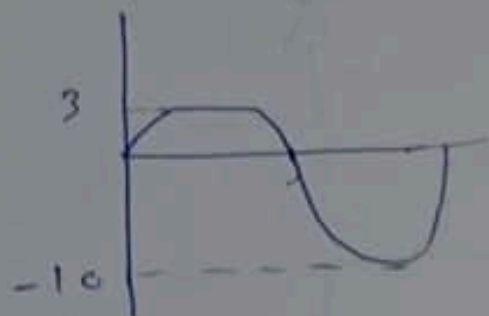
$$-0.7 - 6.2 + V_o = 0$$

$$V_o = 6.9 \text{ volt}$$

$$I_{\text{mer}} = \frac{V_o}{1K} = 6.9 \text{ mA}$$



Q2



$$-10 - 3 + V_o = 0$$

$$V_o = 13$$

Q5

① $-10 + I(1K) + 0.7 = 0$

$I_1 = 19.3 \text{ mA}$

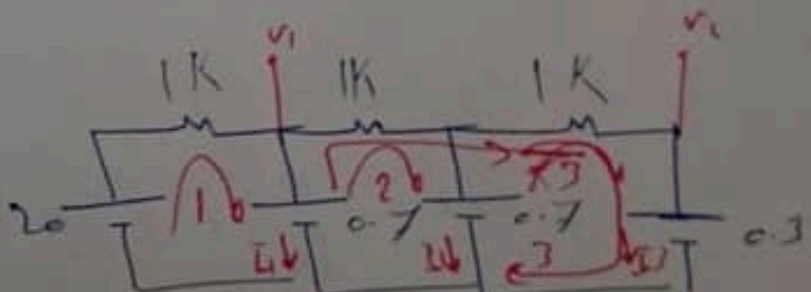
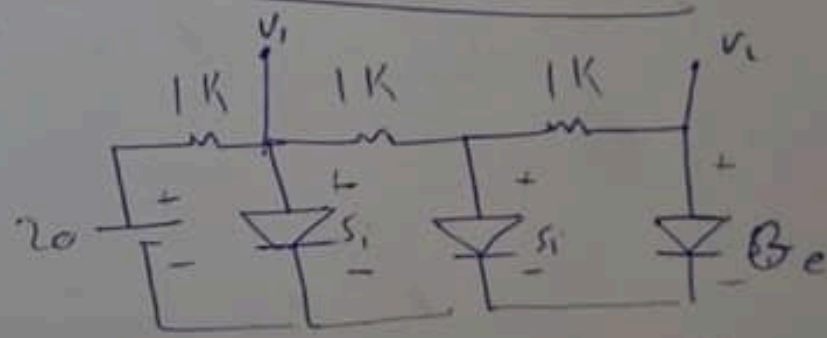
② $-0.7 + I(1K) + 0.7 = 0$

$I_2 = 0 \text{ A}$

③ $-0.7 + I(1K) + 0.7 = 0$

$-0.7 + I(2K) + 0.3 = 0$

$I_3 = 0.2 \text{ mA}$



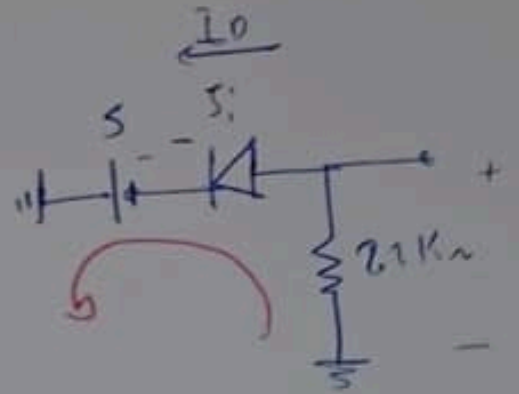
④ $V_1 = V_{D_1} = 0.7 \text{ V}$

Q6

(A)

$$+I_D(2.2K) + 0.7 - 5 = 0$$

$$I_D = 1.95 \text{ mA}$$



(B)

$$(1.95 \text{ mA} \times 2.2K) + V_o = 0$$

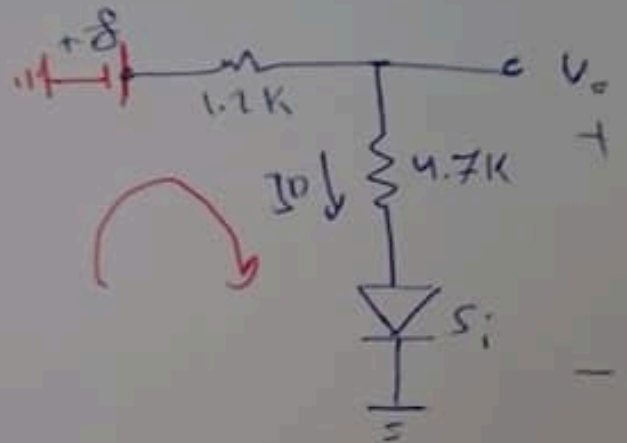
$$V_o = -4.3 \text{ Volt}$$

Q7

(A)

$$-8 + (5.9)K I_D + 0.7 = 0$$

$$I_D = 1.23 \text{ mA}$$



(B)

$$-0.7 - (1.23 \text{ mA})(4.7K) + V_o = 0$$

$$V_o = 6.51 \text{ Volt}$$

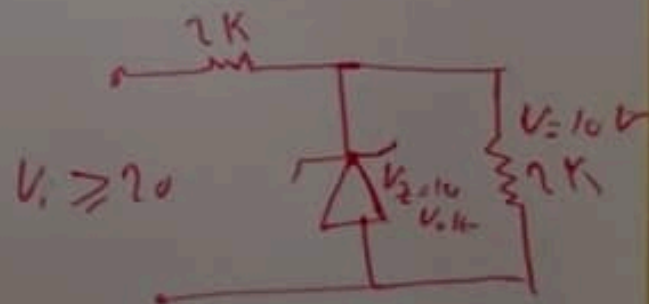
Q8

افرضه في لكقا دقات قد كند $[2K\Omega]$ لكقا و متساوية
 حيث يجب ان تكون (V_i) اكبر من (V_o) .

$$V_o = \frac{V_i R_L}{R_L + R}$$

$$V_i = \frac{V_o (R_L + R)}{R_L} = \frac{10(4K)}{2K}$$

$$V_i = 20 \text{ Volt}$$



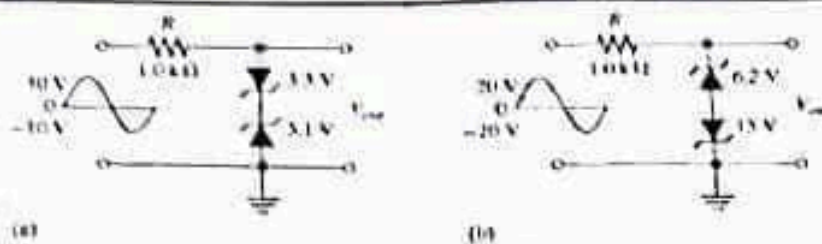


Fig.1

Q.1 A- Referring to figure 1-(a), answer the next questions (assuming that the zener diode in forward acts as a silicon diode):

1. The value of the output voltage at positive half cycle is: _____
2. The maximum current at negative half cycle is: _____

B- Referring to figure 1-(b), answer the next questions (assuming that the zener diode in forward acts as a silicon diode):

1. The value of the output voltage at negative half cycle is: _____
2. The maximum current at positive half cycle is: _____

Q.2 Sketch the output waveform and values, for each case in figures below (assuming that all diodes are ideal):

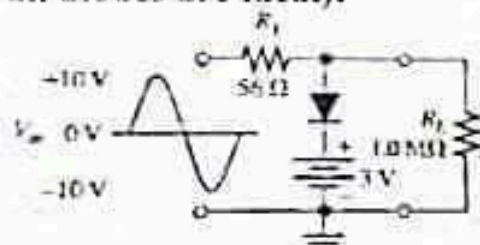


Fig.2

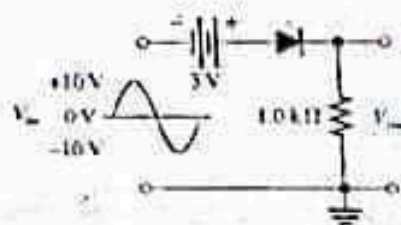


Fig.3

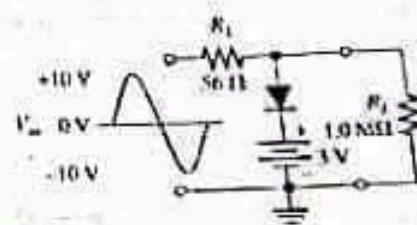


Fig.4

Fig.2

Fig.3

Fig.4



Electronic devices

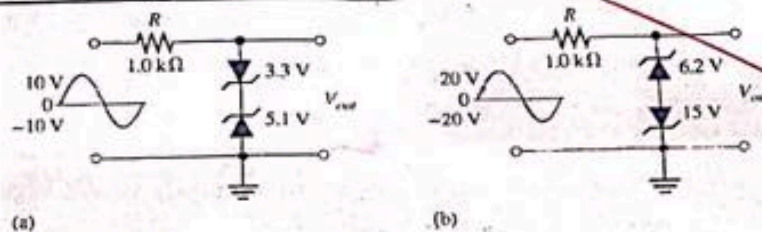


Fig.1

Q.1 A- Referring to figure 1-(a), answer the next questions (assuming that the zener diode in forward acts as a silicon diode):

1. The value of the output voltage at positive half cycle is: 5.8 V
2. The maximum current at negative half cycle is: -4 mA

B- Referring to figure 1-(b), answer the next questions (assuming that the zener diode in forward acts as a silicon diode):

1. The value of the output voltage at negative half cycle is: -15.7 V
2. The maximum current at positive half cycle is: 6.9 mA

Q.2 Sketch the output waveform and values, for each case in figures below (assuming that all diodes are ideal):

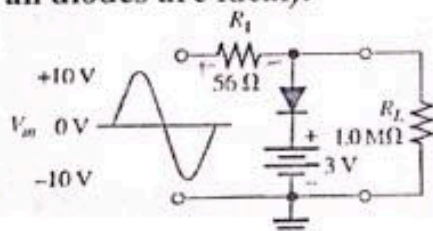


Fig.2

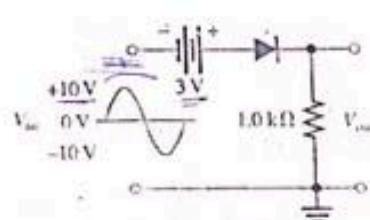


Fig.3

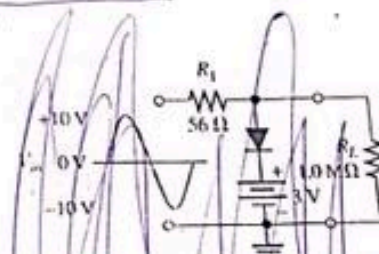


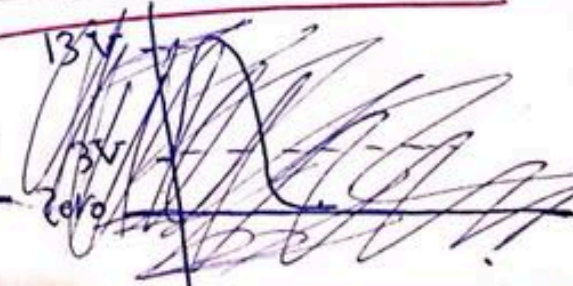
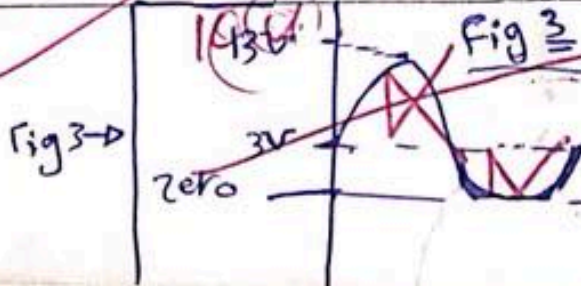
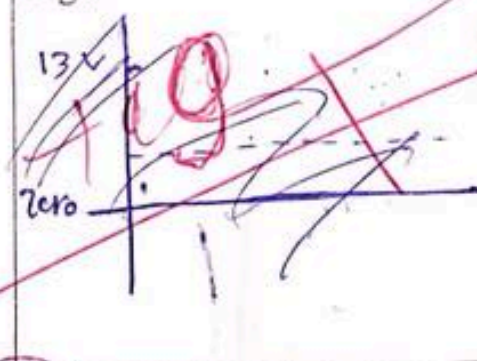
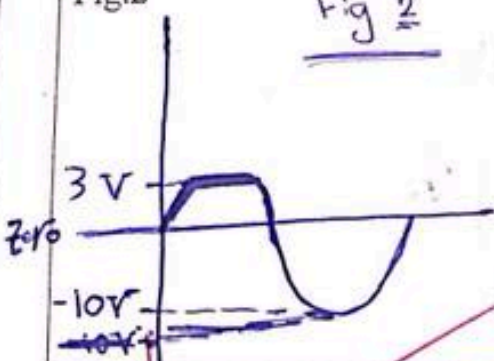
Fig.4

Fig.2

Fig 2

Fig.3

Fig.4



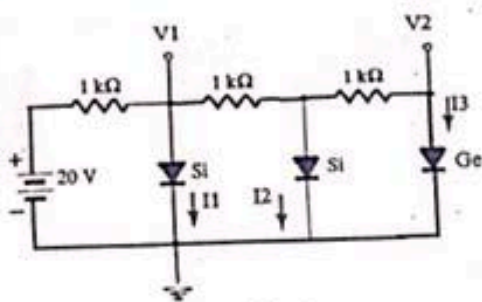


Fig.5

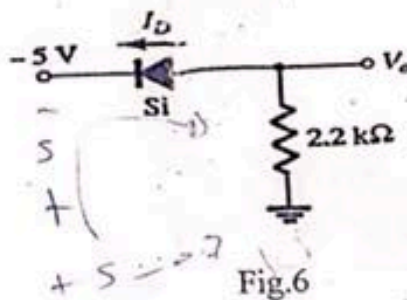


Fig.6

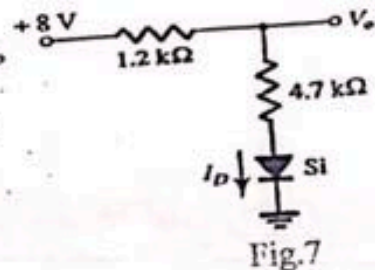


Fig.7

Q.5 With reference to figure 5, answer the next questions:

A- The value of the current I_1 is: 19.1 mA

B- The value of the current I_2 is: zero

C- The value of the current I_3 is: 0.2 mA

D- The value of the voltage V_1 is: 0.7 V

Q.6 With reference to figure 6, answer the next questions:

A- The value of the current I_D is: ~~1.945~~ 1.954545 mA

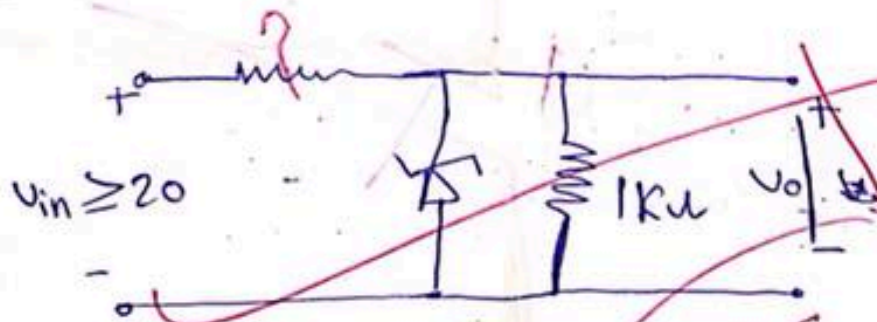
B- The value of the output voltage is: -4.3 V

Q.7 With reference to figure 7, answer the next questions:

A- The value of the current I_D is: 1.2373 mA

B- The value of the output voltage is: 6.615 V

Q.8 Design a voltage regulator circuit, that gives us an output voltage of 10V, choose any suitable values and draw the circuit below with all values on it.



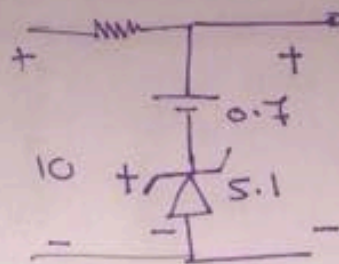
10V

Q1 (A)

①

$$-5.1 - 0.7 + V_o = 0$$

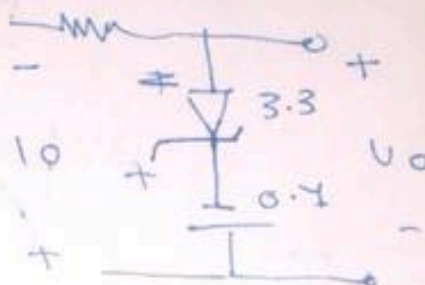
$$\therefore V_o = \boxed{5.8 \text{ V}}$$



②

$$+0.7 + 3.3 + V_o = 0$$

$$\therefore V_o = -4$$



②

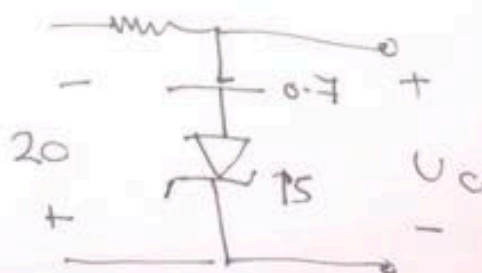
$$\rightarrow I_{max} = \frac{V_o}{1k} = \underline{-4 \text{ mA}}$$

Sultana
abanghan

③

$$15 + 0.7 + V_o = 0$$

$$V_o = -15.7 \text{ V}$$



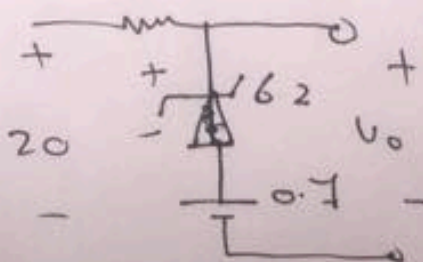
SPARK-team

④

$$-0.7 - 6.2 + V_o = 0$$

$$V_o = 6.9 \text{ V}$$

$$I_{max} = \frac{V_o}{1k} = \underline{6.9 \text{ mA}}$$



Q2

Fig2
↓

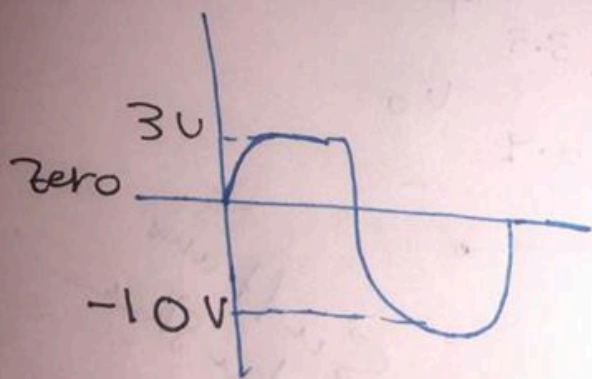
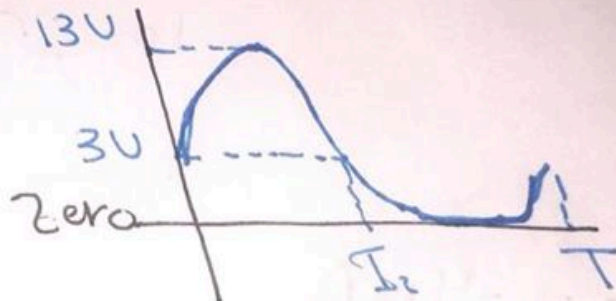


Fig3



Q5

①

$$-20 + I(1k) + 0.7 = 0$$

$$\therefore I_1 = \underline{19.1 \text{ mA}}$$

②

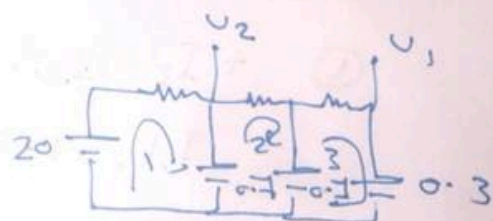
$$-0.7 + I(1k) + 0.7 = 0$$

$$I_2 = \underline{\text{Zero}}$$

③

$$-0.7 + I(2k) + 0.3 = 0$$

$$I_3 = \underline{0.2 \text{ mA}}$$



④

$$V_1 = V_D = \underline{0.7 \text{ Volt}}$$

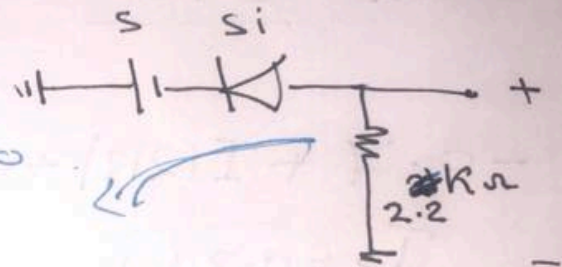
Q8

$$\textcircled{1} +I_D(22k) + 0.7 - 5 = 0$$

$$I_D = 1.9545 \text{ mA}$$

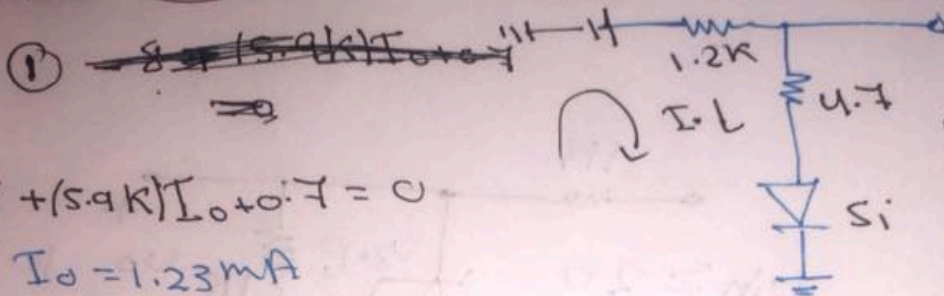
$$\textcircled{2} (1.95 \text{ mA} \times 2.2k) + V_o = 0$$

$$\underline{V_o = -4.3 \text{ V}}$$



spark-team

⑦



$$-8 + (5.9k)I_o + 0.7 = 0$$

$$I_o = 1.23mA$$

$$② \quad -0.7 - (1.23)(4.7k) + V_o = 0$$

$$V_o = 6.815V$$

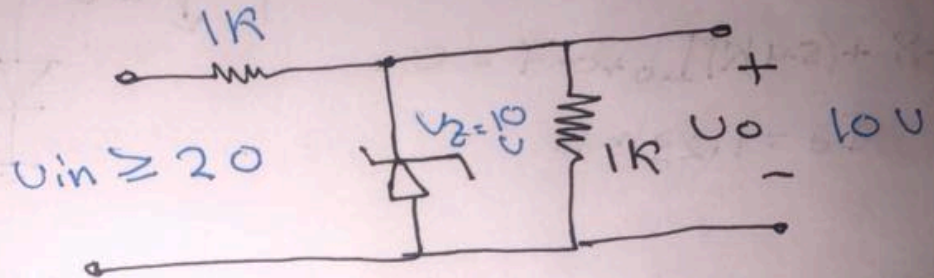
Q 8

$$U_o = \frac{U_i R_L}{R_L + R}$$

$$U_i = \frac{U_o (R_L + R)}{R_L}$$

$$= \frac{10 (4K)}{1K}$$

$$U_i = 20 V$$





Electronic devices



Fig.1



Fig.2

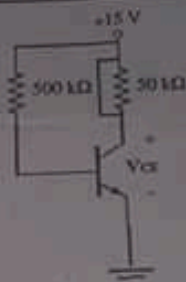


Fig.3

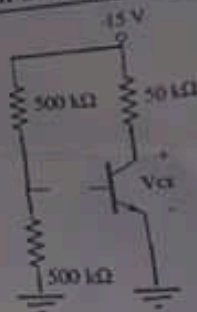


Fig.4

Q.1 Given the information provided in Fig.1, let $\beta = 100$. Then find:

A- The value of the thevenin voltage V_{TH} is: -2.04

B- The value of the thevenin resistance R_{TH} is: 54.68kΩ

C- The value of the current I_{CQ} is: 0.225mA

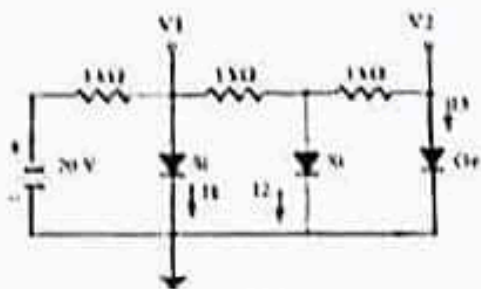


Fig.5

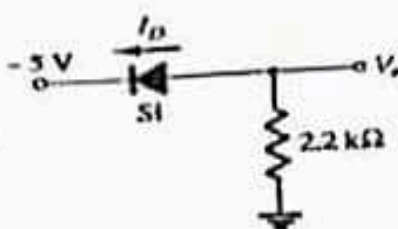


Fig.6

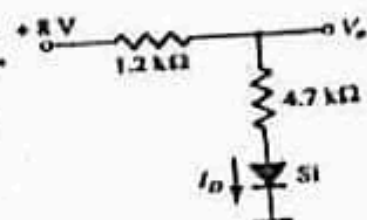


Fig.7

Q.5 With reference to figure 5, answer the next questions:

- A- The value of the current I_1 is: _____
- B- The value of the current I_2 is: _____
- C- The value of the current I_3 is: _____
- D- The value of the voltage V_1 is: _____

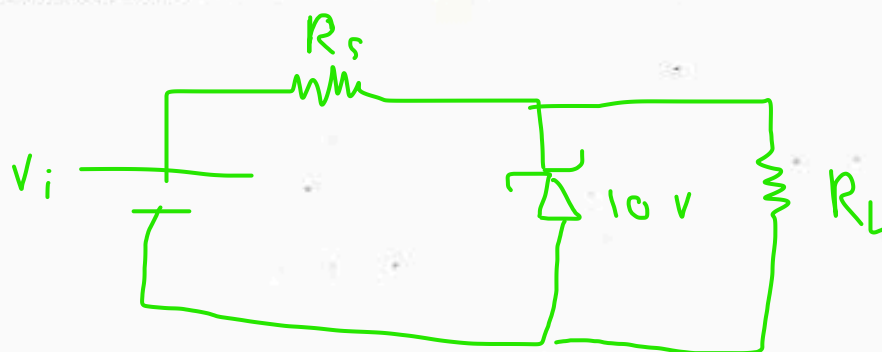
Q.6 With reference to figure 6, answer the next questions:

- A- The value of the current I_1 is: _____
- B- The value of the output voltage is: _____

Q.7 With reference to figure 7, answer the next questions:

- A- The value of the current I_1 is: _____
- B- The value of the output voltage is: _____

Q.8 Design a voltage regulator circuit, that gives us an output voltage of 10v, choose any suitable values and draw the circuit below with all values on it.



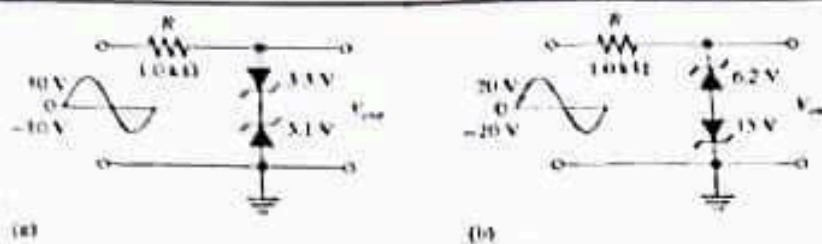


Fig.1

Q.1 A- Referring to figure 1-(a), answer the next questions (assuming that the zener diode in forward acts as a silicon diode):

1. The value of the output voltage at positive half cycle is: _____
2. The maximum current at negative half cycle is: _____

B- Referring to figure 1-(b), answer the next questions (assuming that the zener diode in forward acts as a silicon diode):

1. The value of the output voltage at negative half cycle is: _____
2. The maximum current at positive half cycle is: _____

Q.2 Sketch the output waveform and values, for each case in figures below (assuming that all diodes are ideal):

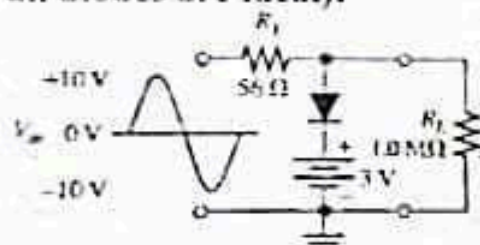


Fig.2

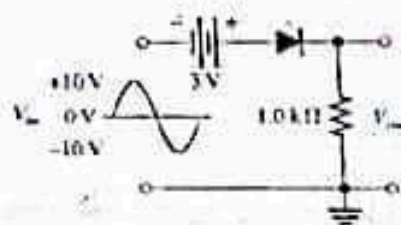


Fig.3

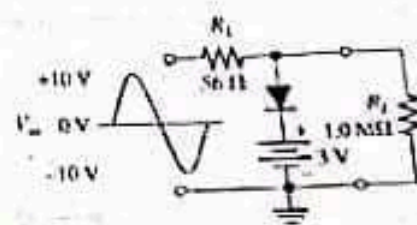


Fig.4

Fig.2

Fig.3

Fig.4

Q.3 with reference to figure 3, answer the next questions (assume $\beta=100$):

- 1- The value of I_C is:
- 2- The value of V_{CE} is:

Q.4 with reference to figure 4, answer the next questions (assume $\beta=100$):

- 1- The value of I_C is:
- 2- The value of V_{CE} is:
- 3- The value of V_E is:

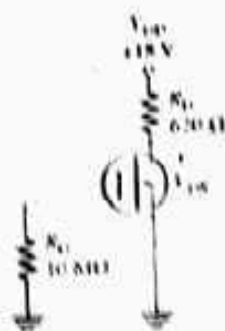


Fig.5

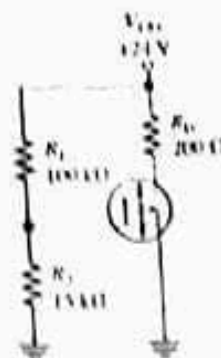


Fig.6

Q.5 with reference to figure 5, answer the next questions ($V_P = -8$, $I_{D(on)} = 12\text{mA}$):

- 1- The value of I_D is:
- 2- The value of V_{DS} is:
- 3- The value of V_{GS} is:

Q.6 with reference to figure 6, answer the next questions ($V_{GS(on)} = 4$, $I_{D(on)} = 200\text{mA}$, and $V_{th} = 2\text{V}$):

- 1- The value of V_{GS} is:
- 2- The value of K is:
- 3- The value of I_D is:
- 4- The value of V_{DS} is:

Q.7 Choose the best answer:

1- The JFET is:

- (a) a unipolar device (b) a current-controlled device (c) a voltage-controlled device
 (d) answers (a) and (c) (e) answers (a) and (b)

2- The channel of a JFET is between the:

- (a) Gate and drain (b) drain and source (c) gate and source (d) input and output



Electronics 1

1 Determine the output voltage waveform for each circuit in Figure 1.

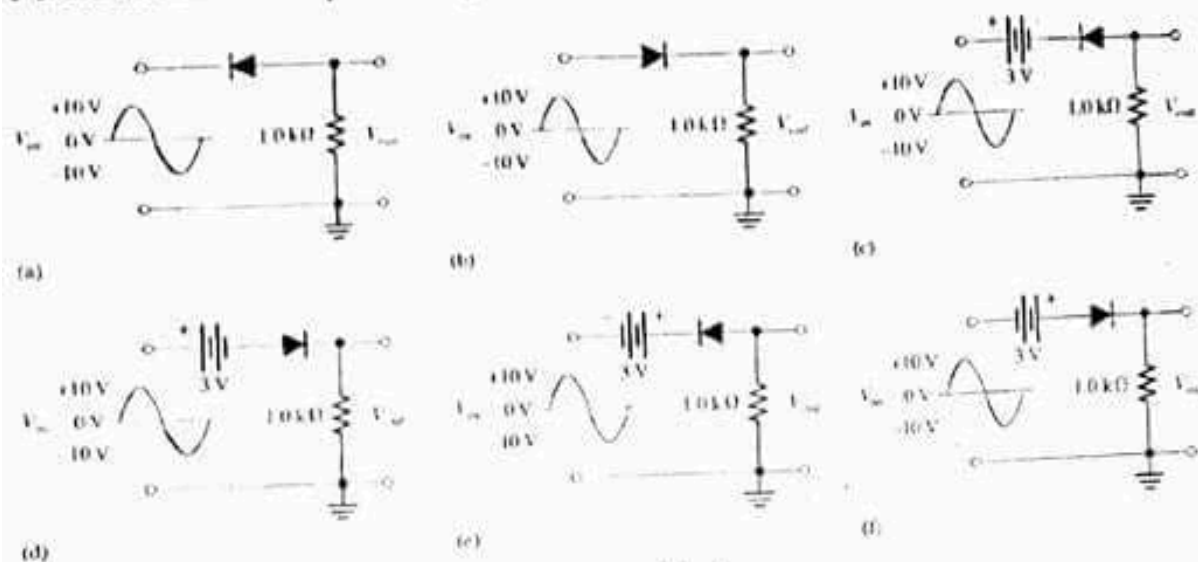


Fig.1

(a)	(b)	(c)
(d)	(e)	(f)

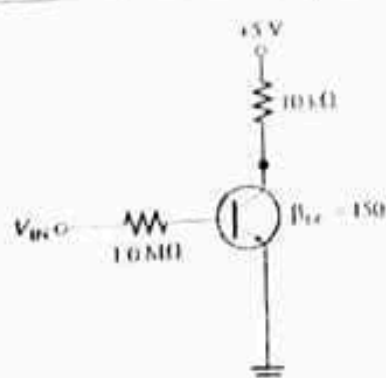


Fig.2

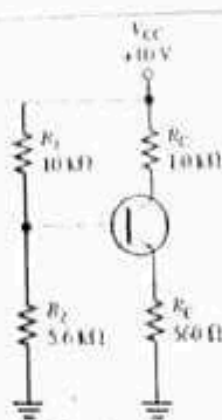


Fig.3

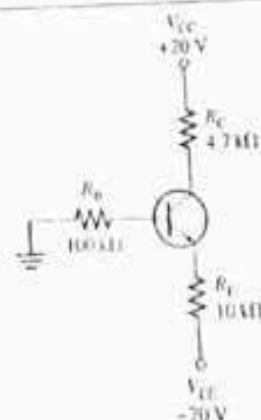


Fig.4

Q.2 with reference to figure 2, answer the next questions:

- 1- The value of I_{Csat} is: _____
- 2- The minimum value of V_{IN} necessary for saturation is: _____

Q.3 with reference to figure 3, answer the next questions (assume $\beta=100$):

- 1- The value of I_C is: _____
- 2- The value of V_{CE} is: _____

Q.4 with reference to figure 4, answer the next questions (assume $\beta=100$):

- 1- The value of I_C is: _____
- 2- The value of V_{CE} is: _____
- 3- The value of V_E is: _____

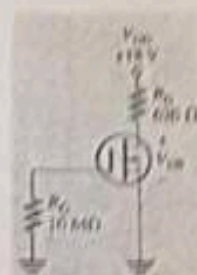


Fig.5

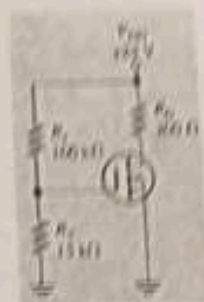


Fig.6

Q.5 with reference to figure 5, answer the next questions ($V_F = -8$, $I_{DS} = 12\text{mA}$):

- 1- The value of I_D is: _____
- 2- The value of V_{DS} is: _____
- 3- The value of V_{GS} is: _____

Q.6 with reference to figure 6, answer the next questions ($V_{GS(on)} = 4$, $I_{D(on)} = 200\text{mA}$, $V_{TH} = 2\text{V}$):

- 1- The value of V_{GS} is: _____
- 2- The value of K is: _____
- 3- The value of I_D is: _____
- 4- The value of V_{DS} is: _____

Q.7 Choose the best answer:

1-The JFET is:

- (a) a unipolar device (b) a current-controlled device (c) a voltage-controlled device
(d) answers (a) and (c) (e) answers (a) and (b)

2- The channel of a JFET is between the:

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- 1- The value of I_C is: _____
- 2- The value of V_{CE} is: _____
- 3- The value of V_E is: _____

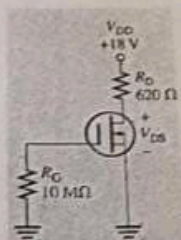


Fig.5

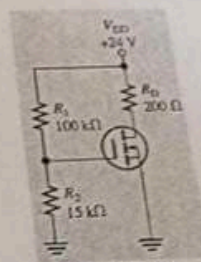


Fig.6

Q.5 with reference to figure 5, answer the next questions ($V_T = -8$, $I_{DSS} = 12\text{mA}$):

- 1- The value of I_D is: _____
- 2- The value of V_{DS} is: _____
- 3- The value of V_{GS} is: _____

Q.6 with reference to figure 6, answer the next questions ($V_{GS(on)} = 4$, $I_{D(on)} = 200\text{mA}$, and $V_{DS(on)} = 2\text{V}$):



[Redacted]

Electronics 1

Q.1 Determine the output voltage waveform for each circuit in Figure 1.

Final exam 16/05/2017

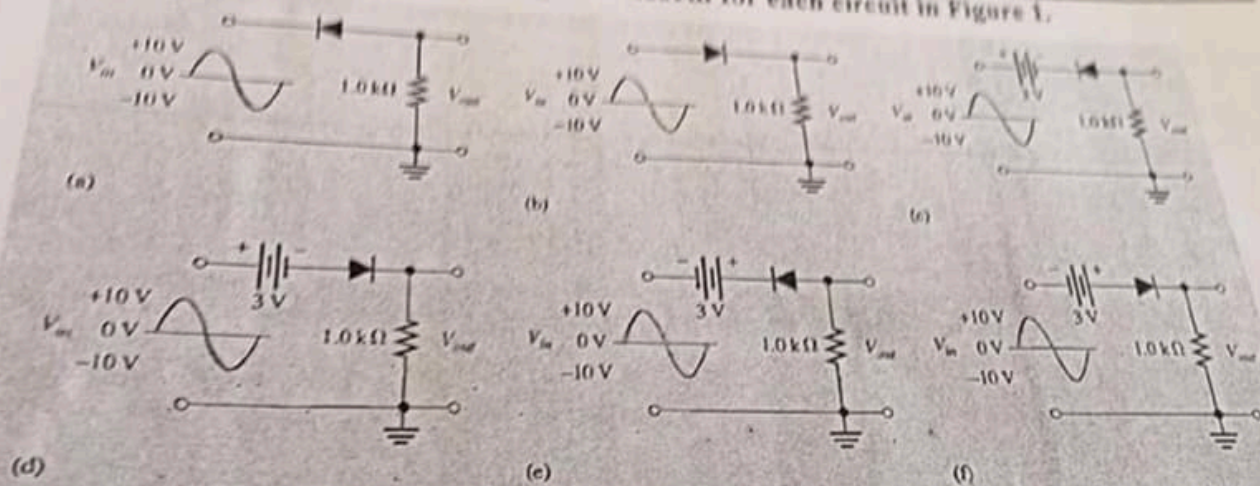


Fig.1

(a)

(b)

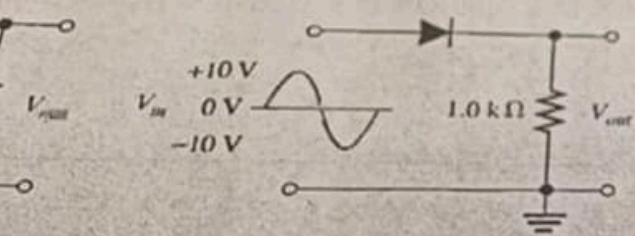
(c)



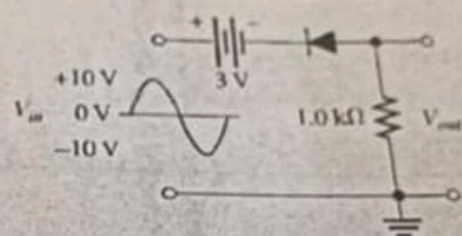
[Redacted Name]

Final Exam 16/05/2017

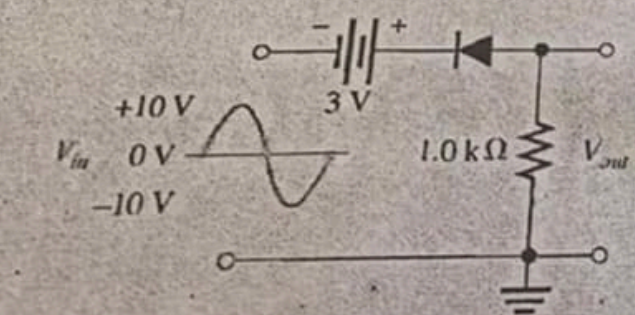
Output voltage waveform for each circuit in Figure 1.



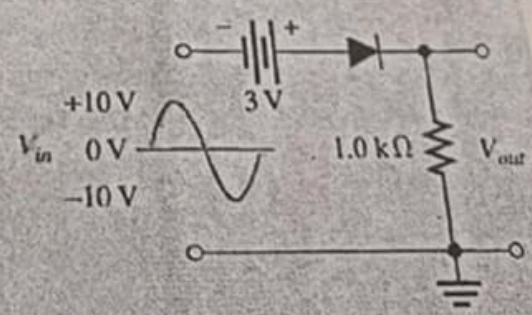
(b)



(c)



(e)



(f)

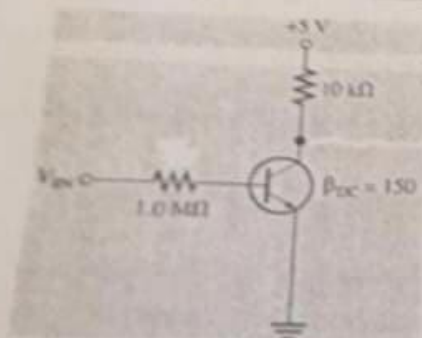


Fig.2

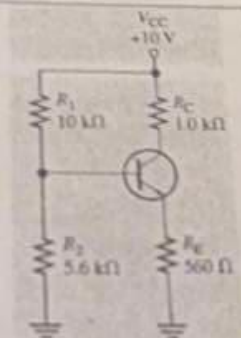


Fig.3

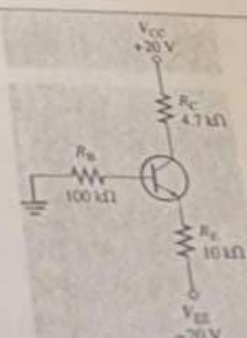


Fig.4

Q.2 with reference to figure 2, answer the next questions:

- 1- The value of $I_{C_{sat}}$ is: _____
- 2- The minimum value of V_{IN} necessary for saturation is: _____

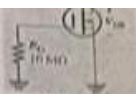


Fig.5

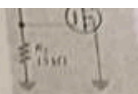


Fig.6

Q.5 with reference to figure 5, answer the next questions ($V_P = -8$, $I_{DSS} = 12\text{mA}$):

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